

Integration

Integration is differentiation backwards: add one to the power, divide by the new power, and never forget +c. Then use a point to find c.



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THE TOPIC ON ONE MAP

Integration

01 - Integrate powers of x

$$\int x^n dx \quad +c$$

02 - Brackets and fractions first

expand divide

03 - Find the curve

the point gives c tangent at P

04 - From $f''(x)$: integrate twice

two constants $f'(2) = 8$

THE ONE RULE

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + c \quad (n \neq -1)$$

power up by one · divide by the new power · +c

01 Integrate Powers of x

Same rewriting as differentiation, then the rule backwards. The number in front is divided by the **new** power.

WHEN YOU SEE → YOU DO

“find $\int \dots dx$ ”, “use calculus to find”

→ power +1, ÷ new power, +c

a number alone, e.g. -5

→ becomes $-5x$

“giving each term in simplest form”

→ simplify every coefficient and power

a fractional power $x^{\frac{1}{2}}$

→ $\div \frac{3}{2}$ means $\times \frac{2}{3}$

WORKED MODEL • JANUARY 2020

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

1 $\frac{8}{3}x^3 - \frac{1}{2}x^{-\frac{1}{2}} - 5$

rewrite

2 $\frac{8}{3} \cdot \frac{x^4}{4} - \frac{1}{2} \cdot \frac{x^{1/2}}{1/2} - 5x$

power up, divide

ANSWER $\frac{2}{3}x^4 - x^{\frac{1}{2}} - 5x + c$

× **TRAP** $\frac{x^{\frac{3}{2}}}{\frac{3}{2}}$ is $\frac{2}{3}x^{\frac{3}{2}}$ — dividing by a fraction multiplies by its flip.

IDEA 01

Practise: Integrate Powers of x

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Power up, divide, +c

P1 • TOPIC 13 3 marks

Find (a) $\int 6x^2 dx$ (b) $\int \frac{4}{x^3} dx$ (c) $\int \sqrt{x} dx$

Q2 DR ESLAM Numbers alone

P1 • TOPIC 13 2 marks

Find $\int (5 - 3x^2) dx$.

SELECTED PRACTICE

Q3 **PRACTICE** Practice · Easy

P1 · TOPIC 13 3 marks

1 Integrate

(i) $2x$,

(ii) $6x^2$,

(iii) $\frac{1}{2}x^{\frac{1}{2}}$.

(3 marks)

Q4 **PRACTICE** Practice · Medium

P1 · TOPIC 13 3 marks

2 Given $y = 2t^2 + t^{-\frac{1}{2}} - 3t^{-4}$, find $\int y \, dt$.

(3 marks)

Q5**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 3 marks

- 1 Use calculus to find

$$\int \left(2\sqrt{x} + 5x^{\frac{1}{3}} \right) dx$$

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q1

P1 · TOPIC 13 4 marks

1. Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)**(Total 4 marks)**

Q7 **PAST PAPER** Pearson IAL WMA11 · October 2021 · Q1

P1 · TOPIC 13 5 marks

1. Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} dx$$

giving each term in simplest form.

(5)

(Total 5 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · January 2022 · Q1

P1 · TOPIC 13 4 marks

1. Find

$$\int \left(\frac{8x^3}{5} - \frac{2}{3x^4} - 1 \right) dx$$

giving each term in simplest form.

(4)

(Total 4 marks)

Q9 **PAST PAPER** Pearson IAL WMA11 · June 2022 · Q1**P1 · TOPIC 13** **4 marks**

1. Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

(Total 4 marks)

Q10 **PAST PAPER** Pearson IAL WMA11 · June 2024 · Q1**P1 · TOPIC 13** **3 marks**

1. Find

$$\int \left(10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

(Total for Question 1 is 3 marks)

Q11 **PAST PAPER** Pearson IAL WMA11 · January 2025 · Q1**P1 · TOPIC 13** **4 marks**

1. Find

$$\int \left(8x^3 - 6\sqrt{x} - \frac{2}{5x^3} \right) dx$$

giving your answer in simplest form.

(4)**(Total for Question 1 is 4 marks)**

Q12 **PAST PAPER** Pearson IAL WMA11 · January 2026 · Q1**P1 · TOPIC 13** **6 marks**

1. $y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \quad x > 0$

Find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\int y \, dx$ (3)

(Total for Question 1 is 6 marks)

02 Brackets and Fractions First

You cannot integrate a product or a quotient term by term. Expand the brackets, split the fraction, then integrate.

WHEN YOU SEE → YOU DO

$$(x-3)^2(2x+5), (3x+\frac{2}{x})^2$$

→ expand first

$$\frac{(x+3)^2}{3\sqrt{x}}$$

→ expand the top, then $\div 3x^{\frac{1}{2}}$ term by term

“given $(x-5)(2x+1)(x+3) \equiv ax^3 + \dots$, hence find”

→ use the expansion you just found

WORKED MODEL • JUNE 2025

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

$$1 \quad \frac{x^2 + 6x + 9}{3x^{\frac{1}{2}}} = \frac{1}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$$

expand, split

$$2 \quad \frac{1}{3} \cdot \frac{2}{5}x^{\frac{5}{2}} + 2 \cdot \frac{2}{3}x^{\frac{3}{2}} + 3 \cdot 2x^{\frac{1}{2}}$$

integrate

$$\text{ANSWER} \quad \frac{2}{15}x^{\frac{5}{2}} + \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + C$$

IDEA 02

Practise: Brackets and Fractions First

My questions first, then selected practice, then real past papers.

Q13 DR ESLAM Expand first

P1 • TOPIC 13 3 marks

Find $\int (x + 2)(x - 5) dx$.

Q14 DR ESLAM Split the fraction

P1 • TOPIC 13 4 marks

Find $\int \frac{x^2 - 4}{\sqrt{x}} dx$.

$x^{\frac{3}{2}} - 4x^{-\frac{1}{2}}$ first.

SELECTED PRACTICE

Q15**PRACTICE**

Practice · Easy

P1 • TOPIC 13

5 marks**3 (a)** Show that

$$\frac{3x^3 + 4x^6}{x^2}$$

can be written as $3x^a + 4x^b$, where a and b are constants to be found.

(2 marks)**(b)** Hence find

$$\int \frac{3x^2 + 4x^6}{x^2} dx.$$

(3 marks)

Q16**PRACTICE**

Practice · Hard

P1 · TOPIC 13 6 marks**4 (a)** Show that

$$(2 - x)^3 = 8 - 12x + 6x^2 - x^3$$

(3 marks)**(b)** Hence, or otherwise, work out

$$\int (2 - x)^3 dx$$

(3 marks)

Q17**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 6 marks**4 (a)** Show that

$$\left(3 - \frac{1}{2}x\right)^3 = 27 - \frac{27}{2}x + \frac{9}{4}x^2 - \frac{1}{8}x^3$$

(3 marks)**(b)** Hence, or otherwise, work out

$$\int \left(2\left(3 - \frac{1}{2}x\right)\right)^3 dx$$

(3 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 • January 2021 • Q9

P1 • TOPIC 13 11 marks

9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}} dx \quad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total 11 marks)

Q19 **PAST PAPER** Pearson IAL WMA11 · June 2021 · Q4**P1 · TOPIC 13** **6 marks**

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(6)

(Total 6 marks)

Q20 **PAST PAPER** Pearson IAL WMA11 • October 2022 • Q2**P1 • TOPIC 13 7 marks**

2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,(a) find the value of a and the value of b .**(2)**

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} \, dx$$

writing each term in simplest form.

(5)**(Total for Question 2 is 7 marks)**

Q21 **PAST PAPER** Pearson IAL WMA11 • January 2023 • Q3**P1 • TOPIC 13** **5 marks**

3. Find

$$\int \frac{4x^5 + 3}{2x^2} dx$$

giving your answer in simplest form.

(5)

(Total for Question 3 is 5 marks)

Q22**PAST PAPER**

Pearson IAL WMA11 • January 2024 • Q1

P1 • TOPIC 13 5 marks

1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)dx$$

writing your answer in simplest form.

(5)**(Total for Question 1 is 5 marks)**

Q23 **PAST PAPER** Pearson IAL WMA11 • June 2025 • Q3**P1 • TOPIC 13** **5 marks**

3. Find

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

writing your answer in simplest form.

(5)**(Total for Question 3 is 5 marks)**

Q24 **PAST PAPER** Pearson IAL WMA11 • October 2025 • Q2**P1 • TOPIC 13** **5 marks**

2. Find

$$\int (x-3)^2(2x+5) \, dx$$

writing the answer in simplest form.

(5)**(Total for Question 2 is 5 marks)**

Q25 **PAST PAPER** Pearson IAL WMA11 • June 2026 • Q1**P1 • TOPIC 13** **5 marks**

1. Find

$$\int \left(3x + \frac{2}{x} \right)^2 dx$$

giving the answer in simplest form.

(5)

Question 1 continued**(Total for Question 1 is 5 marks)**

03 Find the Curve: Use the Point for c

Given $f'(x)$ and one point, integrate to get $f(x)$ with $+c$, then put the point in to find c .

WHEN YOU SEE → YOU DO

“given $f'(x) = \dots$ and $P(4, 20)$ lies on C , find $f(x)$ ”

→ integrate, substitute P , find c

“the curve passes through $(4, 64)$ ”

→ same: the point fixes c

“the gradient at P ” / “tangent at P ”

→ substitute into $f'(x)$ — no integration needed

“tangent at P is $y = 10x + c$ ”

→ gradient = 10; the point is on the line

1

Integrate $f'(x)$

Term by term, with $+c$.

2

Substitute the point

$f(x_0) = y_0$ gives one equation for c .

3

Write $f(x)$ in full

With the number, simplified.

WORKED MODEL • OCTOBER 2022

$$f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4, \quad P(9, 8) \text{ on } C.$$

$$1 \quad f(x) = 24x^{\frac{1}{2}} + \frac{1}{6}x^2 - 4x + c$$

integrate

$$2 \quad 8 = 24(3) + \frac{81}{6} - 36 + c \Rightarrow c = -\frac{83}{2}$$

use P

ANSWER $f(x) = 24\sqrt{x} + \frac{1}{6}x^2 - 4x - \frac{83}{2}$

IDEA 03

Practise: Find the Curve

My questions first, then selected practice, then real past papers.

Q26**DR ESLAM**

One point, one c

P1 · TOPIC 13**4 marks**

$f'(x) = 6x - 4$ and the curve passes through $(2, 5)$. Find $f(x)$.

$f(x) =$ _____

Q27**DR ESLAM**

Tangent given

P1 · TOPIC 13**4 marks**

$f'(x) = 3x^2 + k$. The tangent at $P(1, 4)$ has equation $y = 5x - 1$. Find k and $f(x)$.

SELECTED PRACTICE

Q28**PRACTICE**

Practice · Easy

P1 · TOPIC 13 6 marks

4 (a) Integrate $5x^4 + 6x^2 + 2x + 3$.

(3 marks)

(b) Given that $f(x) = \int (5x^4 + 6x^2 + 2x + 3) dx$ and that the graph of $y = f(x)$ passes through the point $(1, 10)$, find an expression for $f(x)$ in terms of x only.

(3 marks)

Q29**PRACTICE**

Practice · Medium

P1 · TOPIC 13 3 marks

3 Find the equation of the curve passing through the point $(4, 64)$ and given by

$$y = \int (3 - 2x + 6x^2) dx$$

(3 marks)

Q30**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 4 marks

3 Find the equation of the curve passing through the point (4, -8) and given by

$$y = \int \left(\frac{2}{\sqrt{x}} - x - 3 \right) dx$$

(4 marks)

PAST PAPERS

Q31 PAST PAPER Pearson IAL WMA11 · June 2021 · Q6

P1 · TOPIC 13 8 marks

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

- (a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

- (b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

Q32 **PAST PAPER** Pearson IAL WMA11 • October 2021 • Q10**P1 • TOPIC 13 7 marks**

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

Q33 PAST PAPER Pearson IAL WMA11 • January 2022 • Q6

P1 • TOPIC 13 11 marks

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

- (a) (i) find the value of the gradient at P

- (ii) Hence find the equation of the tangent to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

- (b) Find $f(x)$, simplifying your answer.

(7)

(Total 11 marks)

Q34 **PAST PAPER** Pearson IAL WMA11 · June 2022 · Q7**P1 · TOPIC 13** **9 marks**

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

- (a) find the value of A .

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

- (b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

Q35**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q5

P1 · TOPIC 13 9 marks

5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $f(x)$

(5)

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y -axis.

(4)

(Total for Question 5 is 9 marks)

Q36 PAST PAPER Pearson IAL WMA11 • June 2023 • Q8

P1 • TOPIC 13 10 marks

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

- (b) find $f(x)$ giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

Q37 **PAST PAPER** Pearson IAL WMA11 • October 2023 • Q7**P1 • TOPIC 13 10 marks**

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

Q38 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q10**P1 · TOPIC 13** **8 marks**

10. In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

First released on AP - Edexcel Discord
<https://sites.google.com/view/ap-edexcel>

(Total for Question 10 is 8 marks)

Q39 PAST PAPER Pearson IAL WMA11 • June 2024 • Q10

P1 • TOPIC 13 10 marks

10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $f(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

Q40 **PAST PAPER** Pearson IAL WMA11 • October 2024 • Q8**P1 • TOPIC 13 7 marks**

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $f(x)$.

(4)

(Total for Question 8 is 7 marks)

Q41 PAST PAPER Pearson IAL WMA11 • January 2025 • Q6

P1 • TOPIC 13 10 marks

6. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24 \quad (2)$$

Given also that $a + b = -9$

(b) find, in simplest form, $f(x)$ (7)

Curve C is transformed to the curve with equation $y = f(x - 3)$

Given that point P is transformed to the point Q ,

(c) state the coordinates of Q . (1)

(Total for Question 6 is 10 marks)

Q42 **PAST PAPER** Pearson IAL WMA11 • June 2025 • Q8**P1 • TOPIC 13** **11 marks**

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = f(x)$, $x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P .

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

Q43**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q8

P1 · TOPIC 13 8 marks

8. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k .

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

Q44 PAST PAPER Pearson IAL WMA11 • January 2026 • Q8

P1 • TOPIC 13 12 marks

8. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

- (a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P ,
(ii) find, using algebra, the other possible x coordinate of P .

(6)

Given that the x coordinate of P is $\sqrt{3}$

- (b) find an equation of the normal to C at P .

(2)

- (c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

Q45 **PAST PAPER** Pearson IAL WMA11 • June 2026 • Q8**P1 • TOPIC 13** **11 marks**

8. A curve C has equation $y = f(x)$, $x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C .

Given that

- $f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(a) (i) find the value of k

(ii) find the equation of the normal to C at P .

Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

(b) Find $f(x)$.

(5)

Question 8 continued

(Total for Question 8 is 11 marks)

04 From $f''(x)$: Integrate Twice

Two integrations, two constants — so you need two facts: a gradient and a point.

WHEN YOU SEE → YOU DO

$$“f''(x) = \dots”, “\frac{d^2y}{dx^2} = \dots”$$

→ integrate twice

$$“f'(x) = 5 \text{ at } P”, “f'(2) = 8”$$

→ fixes the **first** constant

$$“P \text{ lies on the curve}”, “f(3) = 20”$$

→ fixes the **second** constant

$$“f''(x) = 0 \text{ when } x = 27”, “\text{find } a / k”$$

→ differentiate the given f' , set = 0

$$“(2x - 1) \text{ and } (3x + 2) \text{ are factors of } f(x)”$$

→ $f(\frac{1}{2}) = 0$ and $f(-\frac{2}{3}) = 0$

WORKED MODEL • JANUARY 2020

$$f''(x) = \frac{6}{\sqrt{x^3}} + x, \quad f'(4) = -4, \quad P(4, -50).$$

$$1 \quad f'(x) = -12x^{-\frac{1}{2}} + \frac{1}{2}x^2 + c_1; \quad -4 = -6 + 8 + c_1 \Rightarrow c_1 = -6$$

first integration

$$2 \quad f(x) = -24x^{\frac{1}{2}} + \frac{1}{6}x^3 - 6x + c_2$$

second integration

$$3 \quad -50 = -48 + \frac{32}{3} - 24 + c_2 \Rightarrow c_2 = \frac{34}{3}$$

use P

$$\text{ANSWER} \quad f(x) = \frac{1}{6}x^3 - 24\sqrt{x} - 6x + \frac{34}{3}$$

IDEA 04

Practise: From $f''(x)$: Integrate Twice

My questions first, then selected practice, then real past papers.

Q46 DR ESLAM Integrate twice

P1 · TOPIC 13 5 marks

$f''(x) = 12x$, $f'(1) = 10$ and $f(1) = 3$. Find $f(x)$.

$f(x) =$ _____

SELECTED PRACTICE

Q47

PRACTICE

Practice · Medium

P1 · TOPIC 13 7 marks

5 (a) Given $\frac{d^2 y}{dx^2} = 30x$ and that when $x = 1$, $\frac{dy}{dx} = 17$, show that

$$\frac{dy}{dx} = 15x^2 + 2$$

(4 marks)

(b) Find an equation for y in terms of x , given that when $x = 2$, $y = 40$.

(3 marks)

Q48**PRACTICE**

Practice · Hard

P1 · TOPIC 13**5 marks**

5 A function, $f(x)$, has second derivative given by

$$f''(x) = 6(x - 2).$$

Given that $f(3) = 20$, and $f'(2) = 8$, find $f(x)$.

(5 marks)

Q49**PRACTICE**

Practice · Very Hard

P1 · TOPIC 13 6 marks

5 A function, $f(x)$, has second derivative given by

$$f''(x) = 2(18x - 5).$$

Given that $(2x - 1)$ and $(3x + 2)$ are factors of $f(x)$, find $f(x)$.

(6 marks)

PAST PAPERS

Q50 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q11**P1 · TOPIC 13** **10 marks**

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

Q51 **PAST PAPER** Pearson IAL WMA11 • January 2020 • Q11**P1 • TOPIC 13** **11 marks**

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

- (b) find $f(x)$.

(8)

(Total 11 marks)

Q52 **PAST PAPER** Pearson IAL WMA11 • January 2023 • Q11**P1 • TOPIC 13** **8 marks**

11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------------|--|
| 01 | The rule | Power +1, $\div$ new power, +c. Rewrite roots and fractions first. |
| 02 | Brackets / fractions | Expand, split, then integrate term by term. |
| 03 | Find the curve | Integrate $f'(x)$, the point gives c. |
| 04 | From $f''(x)$ | Integrate twice: gradient fact $\rightarrow c_1$, point $\rightarrow c_2$. |

BOTTOM LINE Rewrite $\rightarrow$ power up, divide $\rightarrow$ +c $\rightarrow$ a point finds c; a gradient finds the c of f' .

Answers

Integration • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • INTEGRATE POWERS OF X

- Q1** Dr Eslam • Power up, divide, +c (a) $2x^3 + c$ (b) $-2x^{-2} + c$ (c) $\frac{2}{3}x^{\frac{3}{2}} + c$
- Q2** Dr Eslam • Numbers alone $5x - x^3 + c$
- Q3** Practice • Easy (i) $x^2 + c$ (ii) $2x^3 + c$ (iii) $\frac{1}{3}x^{\frac{3}{2}} + c$
- Q4** Practice • Medium $\frac{2}{3}t^3 + 2t^{\frac{1}{2}} + t^{-3} + c$
- Q5** Practice • Very Hard $\frac{4}{3}x^{\frac{3}{2}} + \frac{15}{4}x^{\frac{4}{3}} + c$
- Q6** Pearson IAL WMA11 • January 2020 • Q1 $\frac{2}{3}x^4 - x^{\frac{1}{2}} - 5x + c$
- Q7** Pearson IAL WMA11 • October 2021 • Q1 $3x^4 + \frac{1}{3}x^{\frac{1}{2}} + \frac{1}{2}x^{-3} + c$
- Q8** Pearson IAL WMA11 • January 2022 • Q1 $\frac{2}{5}x^4 + \frac{2}{9}x^{-3} - x + c$
- Q9** Pearson IAL WMA11 • June 2022 • Q1 $\frac{5}{3}x^6 + \frac{3}{2}x^4 + 3x^{-1} + c$
- Q10** Pearson IAL WMA11 • June 2024 • Q1 $2x^5 + \frac{3}{4}x^{-2} - 7x + c$
- Q11** Pearson IAL WMA11 • January 2025 • Q1 $2x^4 - 4x^{\frac{3}{2}} + \frac{1}{5}x^{-2} + c$
- Q12** Pearson IAL WMA11 • January 2026 • Q1 (a) $12x^2 - \frac{3}{4}x^{-\frac{3}{2}}$ (b) $x^4 + 3x^{\frac{1}{2}} - 8x + c$

IDEA 02 • BRACKETS AND FRACTIONS FIRST

- Q13** Dr Eslam • Expand first $\frac{1}{3}x^3 - \frac{3}{2}x^2 - 10x + c$
- Q14** Dr Eslam • Split the fraction $\frac{2}{5}x^{\frac{5}{2}} - 8x^{\frac{1}{2}} + c$
- Q15** Practice • Easy (a) $a = 1, b = 4$ (b) $3x + \frac{4}{5}x^5 + c$
- Q16** Practice • Hard (b) $8x - 6x^2 + 2x^3 - \frac{1}{4}x^4 + c$
- Q17** Practice • Very Hard (b) $216x - 54x^2 + 6x^3 - \frac{1}{4}x^4 + c$
- Q18** Pearson IAL WMA11 • January 2021 • Q9 (i) $\frac{9}{10}x^{\frac{5}{2}} + 2x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + c$ (ii) $f(x) = \frac{1}{3}x^3 - 2x^2 + 5x - 8$
- Q19** Pearson IAL WMA11 • June 2021 • Q4 $\frac{3}{8}x^2 + \frac{1}{3}x^{\frac{3}{2}} - \frac{15}{4}x - 5x^{\frac{1}{2}} + c$
- Q20** Pearson IAL WMA11 • October 2022 • Q2 (a) $a = 2, b = -3$ (b) $\frac{4}{35}x^{\frac{7}{2}} - \frac{6}{25}x^{\frac{5}{2}} - \frac{64}{15}x^{\frac{3}{2}} - 6x^{\frac{1}{2}} + c$
- Q21** Pearson IAL WMA11 • January 2023 • Q3 $\frac{1}{2}x^4 - \frac{3}{2}x^{-1} + c$
- Q22** Pearson IAL WMA11 • January 2024 • Q1 $3x^4 + \frac{8}{3}x^3 - \frac{75}{2}x^2 - 50x + c$
- Q23** Pearson IAL WMA11 • June 2025 • Q3 $\frac{2}{15}x^{\frac{5}{2}} + \frac{4}{3}x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + c$
- Q24** Pearson IAL WMA11 • October 2025 • Q2 $\frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c$
- Q25** Pearson IAL WMA11 • June 2026 • Q1 $3x^3 + 12x - \frac{4}{x} + c$

IDEA 03 • FIND THE CURVE

- Q26** Dr Eslam • One point, one c $3x^2 - 4x + 1$
- Q27** Dr Eslam • Tangent given $k = 2; f(x) = x^3 + 2x + 1$
- Q28** Practice • Easy (a) $x^5 + 2x^3 + x^2 + 3x + c$ (b) $f(x) = x^5 + 2x^3 + x^2 + 3x + 3$
- Q29** Practice • Medium $y = 2x^3 - x^2 + 3x - 60$
- Q30** Practice • Very Hard $y = 4\sqrt{x} - \frac{1}{2}x^2 - 3x + 4$
- Q31** Pearson IAL WMA11 • June 2021 • Q6 (a) $y = -\frac{5}{6}x + \frac{26}{3}$ (b) $f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{\frac{4}{3}} + \frac{10}{3}$
- Q32** Pearson IAL WMA11 • October 2021 • Q10 (a) $a = \frac{4}{9}$ (b) $f(x) = \frac{2}{9}x^2 - 9x^{\frac{4}{3}} + \frac{7}{9}$
- Q33** Pearson IAL WMA11 • January 2022 • Q6 (a) $\frac{49}{8}; 49x - 8y - 36 = 0$ (b) $f(x) = \frac{2}{3}x^{\frac{3}{2}} + 12x^{\frac{1}{2}} - 18x^{-\frac{1}{2}} - \frac{1}{3}$
- Q34** Pearson IAL WMA11 • June 2022 • Q7 (a) $A = -4$ (b) $f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$
- Q35** Pearson IAL WMA11 • October 2022 • Q5 (a) $f(x) = 24\sqrt{x} + \frac{1}{6}x^2 - 4x - \frac{83}{2}$ (b) $(0, 11)$
- Q36** Pearson IAL WMA11 • June 2023 • Q8 (a) $12x - y - 36 = 0$ (b) $f(x) = \frac{1}{16}x^4 - 16x^{\frac{1}{2}} + 28$
- Q37** Pearson IAL WMA11 • October 2023 • Q7 (a) $\frac{15}{2}; 2x + 15y + 7 = 0$ (b) $f(x) = \frac{2}{5}x^{\frac{5}{2}} - \frac{7}{4}x + 5x^{\frac{1}{2}} - \frac{84}{5}$
- Q38** Pearson IAL WMA11 • January 2024 • Q10 (a) $k = 24\sqrt{2}$ (b) $f(x) = \frac{8}{5}x^{\frac{5}{2}} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}$
- Q39** Pearson IAL WMA11 • June 2024 • Q10 (a) $y = 2x + 4$ (b) $f(x) = 3x^2 - \frac{6}{5}x^{\frac{5}{2}} - \frac{1}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + \frac{16}{15}$
- Q40** Pearson IAL WMA11 • October 2024 • Q8 (b) $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}$
- Q41** Pearson IAL WMA11 • January 2025 • Q6 (b) $a = 11, b = -20: f(x) = \frac{1}{5}x^{\frac{5}{2}} + \frac{11}{6}x^{\frac{3}{2}} - 10x^{\frac{1}{2}} - \frac{91}{15}$ (c) $Q(7, -5)$
- Q42** Pearson IAL WMA11 • June 2025 (a) $k = \frac{11}{4}, c = -28$ (b) $\frac{11}{4}$ (c) $f(x) = 2x^{\frac{3}{2}} + \frac{1}{12}x^3 - \frac{28}{3}$

IDEA 04 • FROM $f''(x)$: INTEGRATE TWICE

- Q46** Dr Eslam • Integrate twice $2x^3 + 4x - 3$
- Q47** Practice • Medium (b) $y = 5x^3 + 2x - 4$
- Q48** Practice • Hard $f(x) = x^3 - 6x^2 + 20x - 13$
- Q49** Practice • Very Hard $f(x) = 6x^3 - 5x^2 - 3x + 2$
- Q50** Pearson IAL WMA11 • October 2019 • Q11 (a) $y = 5x - \frac{28}{3}$ (b) $f(x) = \frac{16}{3}x^{\frac{3}{2}} - \frac{3}{2}x^2 + x - 12$
- Q51** Pearson IAL WMA11 • January 2020 • Q11 (a) $y = \frac{1}{4}x - 51$ (b) $f(x) = \frac{1}{6}x^3 - 24\sqrt{x} - 6x + \frac{34}{3}$
- Q52** Pearson IAL WMA11 • January 2023 • Q11 (a) $y = -\frac{1}{3}x + \frac{52}{3}$ (b) $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{\frac{3}{2}} - 33x + \frac{284}{3}$